AWS and Azure Cloud Pricing Comparison Report

Small Web Application Hosting Cost Evaluation

1. Introduction

This report compares the estimated monthly cost of hosting a small web application on Amazon Web Services (AWS) and Microsoft Azure. The comparison covers compute, storage, networking costs, cloud discount mechanisms, and cost optimization strategies.

2. Application Specifications

Component	Specification
Application type	Small web application
Compute	1 virtual machine
CPU	2 vCPU
Memory	4 GB RAM
VM storage	50 GB
Object storage	100 GB
Monthly data transfer	100 GB outbound
Deployment	Single availability zone

3. AWS Monthly Cost Estimate

The AWS Pricing Calculator estimate includes Amazon EC2 for compute and Amazon S3 Standard for object storage.

AWS Service	Description	Monthly Cost
Amazon EC2	Linux t3.medium compute estimate	$15.04
Amazon S3	100 GB S3 Standard storage estimate	$2.30
Total	AWS estimated monthly cost	$17.34

Figure 1: AWS EC2 estimate

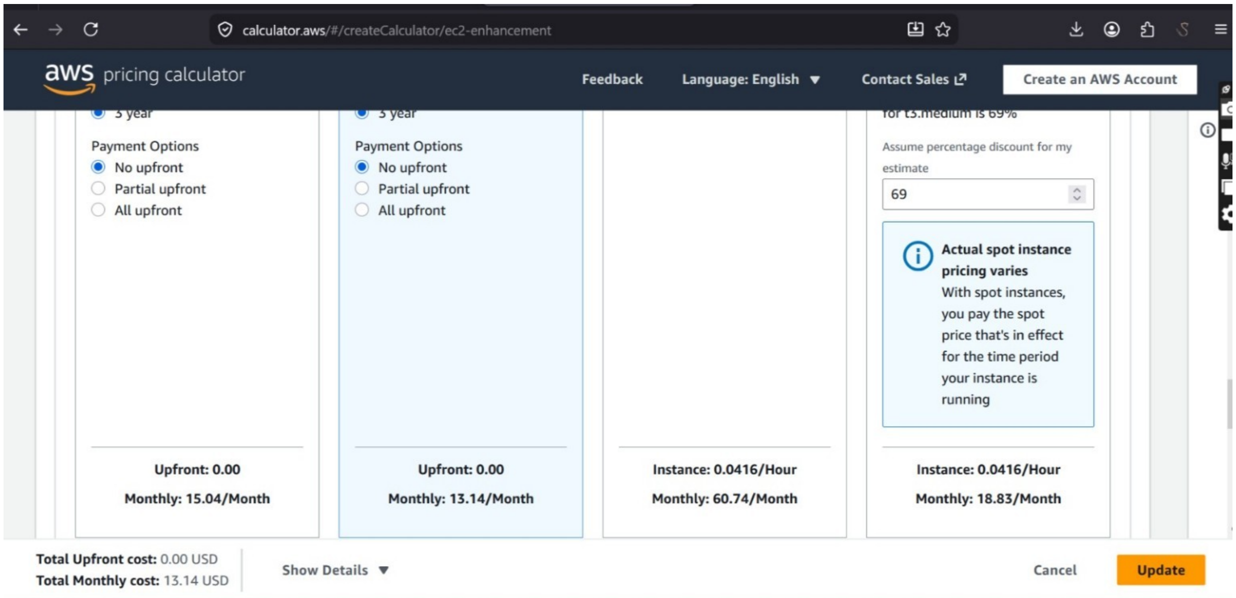


Figure 2: AWS S3 storage estimate

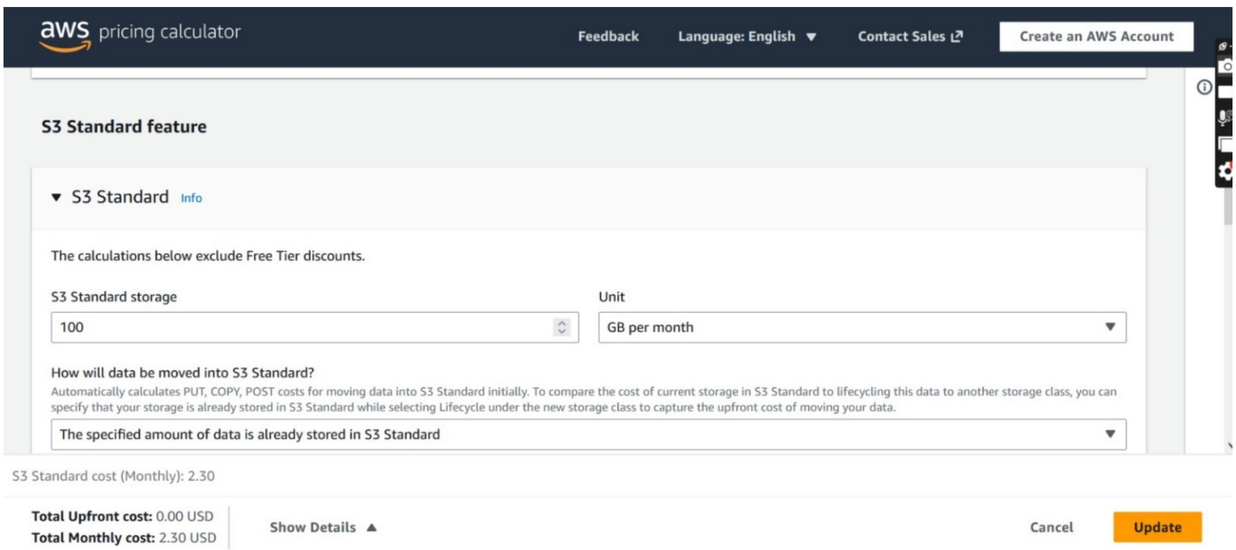
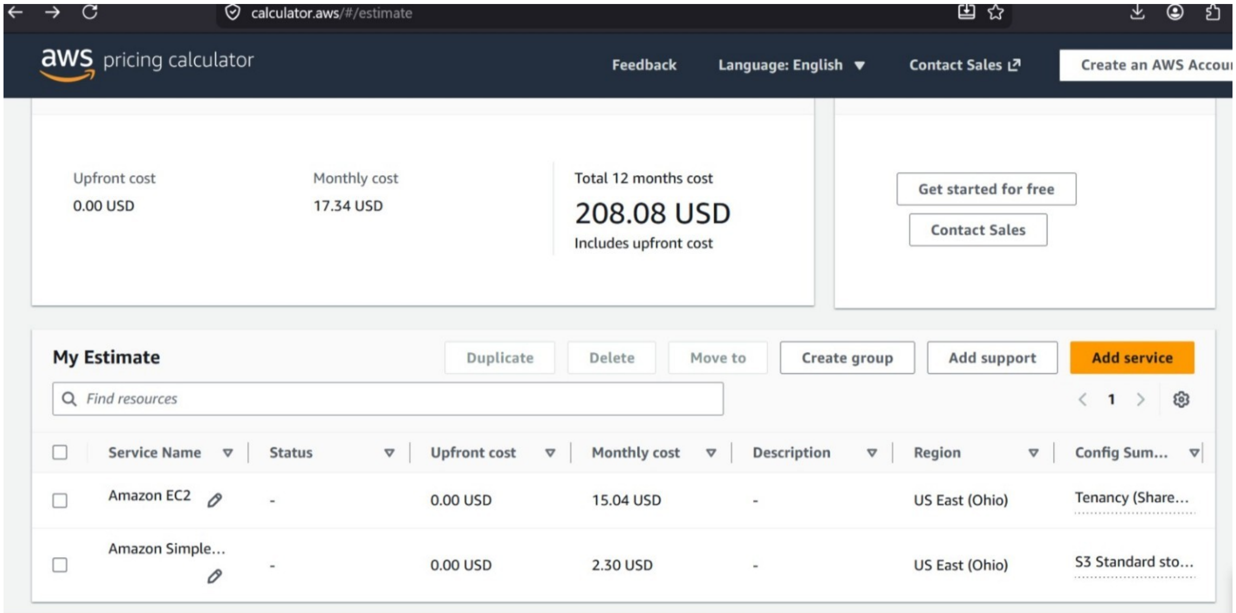


Figure 3: AWS total estimate

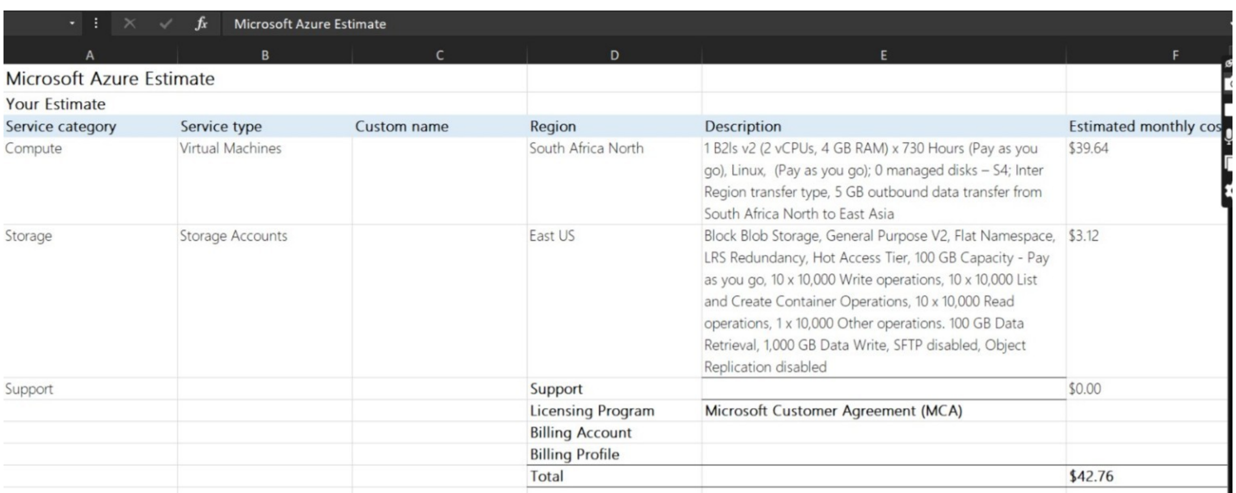


4. Azure Monthly Cost Estimate

The Azure Pricing Calculator estimate includes Azure Virtual Machines and Azure Blob Storage. Both Linux and Windows scenarios were considered.

Azure Scenario	Description	Monthly Cost
Linux VM + Storage	B2ls v2, 2 vCPU, 4 GB RAM, 100 GB storage	$42.76
Windows VM + Storage	B2ls v2, Windows license included, 100 GB storage	$49.48

Figure 4: Azure Linux estimate



Microsoft Azure Estimate

Your Estimate

Service category	Service type	Custom name	Region	Description	Estimated monthly cos
Compute	Virtual Machines		South Africa North	1 B2ls v2 (2 vCPUs, 4 GB RAM) x 730 Hours (Pay as you go), Linux, (Pay as you go); 0 managed disks – S4; Inter Region transfer type, 5 GB outbound data transfer from South Africa North to East Asia	$39.64
Storage	Storage Accounts		East US	Block Blob Storage, General Purpose V2, Flat Namespace, LRS Redundancy, Hot Access Tier, 100 GB Capacity - Pay as you go, 10 x 10,000 Write operations, 10 x 10,000 List and Create Container Operations, 10 x 10,000 Read operations, 1 x 10,000 Other operations. 100 GB Data Retrieval, 1,000 GB Data Write, SFTP disabled, Object Replication disabled	$3.12
Support			Support		$0.00
			Licensing Program	Microsoft Customer Agreement (MCA)	
			Billing Account		
			Billing Profile		
			Total		$42.76

Figure 5: Azure Windows estimate

Microsoft Azure Estimate					
Your Estimate					
Service category	Service type	Custom name	Region	Description	Estimated monthly cos
Compute	Virtual Machines		South Africa North	1 B2ls v2 (2 vCPUs, 4 GB RAM) x 730 Hours (Pay as you go), Windows (License included), OS Only; 0 managed disks – S4; Inter Region transfer type, 5 GB outbound data transfer from South Africa North to East Asia	$46.36
Storage	Storage Accounts		East US	Block Blob Storage, General Purpose V2, Flat Namespace, LRS Redundancy, Hot Access Tier, 100 GB Capacity - Pay as you go, 10 x 10,000 Write operations, 10 x 10,000 List and Create Container Operations, 10 x 10,000 Read operations, 1 x 10,000 Other operations. 100 GB Data Retrieval, 1,000 GB Data Write, SFTP disabled, Object Replication disabled	$3.12
Support			Support		$0.00
			Licensing Program	Microsoft Customer Agreement (MCA)	
			Billing Account		
			Billing Profile		
			Total		$49.48

5. Networking Cost Comparison

Both AWS and Azure charge for inter-zone data transfer. AWS shows approximately $0.01 per GB in each direction across Availability Zones, while Azure shows approximately $0.01 per GB for inter-zone data transfer. Since this deployment uses a single availability zone, inter-zone transfer costs are expected to be minimal.

Figure 6: AWS inter-zone transfer note

AWS charges **$0.01 per GB in each direction** for data transferred across **Availability Zones (AZs)** within the same AWS region. Amazon Web Services (AWS) +1

When moving data between AZs (e.g., across VPC subnets or between EC2 instances and managed databases), you pay both for sending and receiving the data, effectively making it **$0.02 per GB round-trip**. Medium · Ismail Kovvuru +1

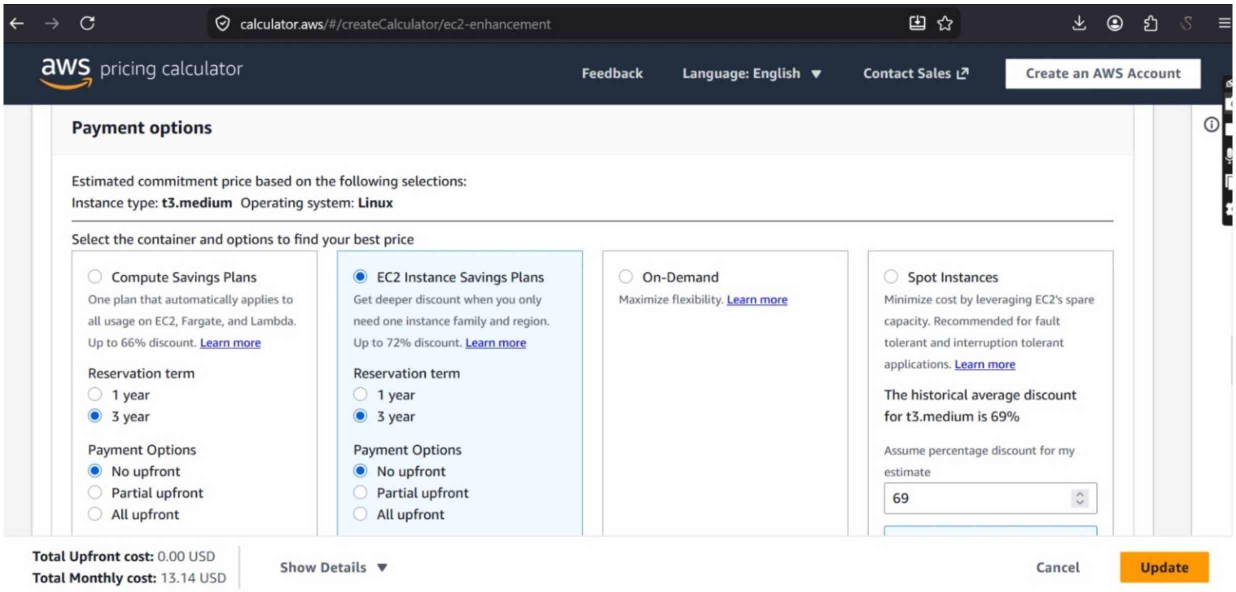
Figure 7: Azure inter-zone transfer note

Data transfer within the same Azure Availability Zone is free. However, inter-zone data transfer—moving data between different Availability Zones in the same region—costs **$0.01 per GB** for both ingress and egress traffic. Microsoft Community Hub +3

6. Discount Mechanisms

AWS provides Savings Plans that reduce compute costs when users commit to consistent usage for one or three years. Azure provides Reserved Virtual Machine Instances and Azure Hybrid Benefit, which can reduce VM costs, especially for Windows workloads when eligible Microsoft licenses are available.

Figure 8: AWS Savings Plan options



7. Final Cost Comparison

Provider	Scenario	Estimated Monthly Cost	Observation
AWS	Linux EC2 + S3	$17.34	Lowest estimated monthly cost
Azure	Linux VM + Storage	$42.76	Higher than AWS for this scenario
Azure	Windows VM + Storage	$49.48	Highest cost due to Windows licensing

8. Recommendation

AWS is the most cost-effective option for this small Linux-based web application based on the estimates provided. Azure may be more suitable for organizations already using Microsoft technologies or where Windows licensing benefits can be applied.

9. Cost Optimization Strategies

1. Use AWS Savings Plans or Azure Reserved Virtual Machine Instances for predictable long-term workloads.

2. Right-size virtual machines and keep services within one availability zone where possible to reduce unnecessary compute and networking costs.

10. Conclusion

This comparison shows that both AWS and Azure can host a small web application, but AWS provides the lowest monthly estimate in this specific scenario. The final provider choice should consider cost, existing licenses, technical requirements, and long-term cloud strategy.